

Benedict Leung, MSc

HUMAN-COMPUTER INTERACTION RESEARCHER

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HCI researcher with an MSc in Computer Science, specializing in novel interaction techniques, multimodal sensing, and human-AI interaction. Builds end-to-end interactive systems using eye tracking, IMUs, pen/stylus input, and brain-computer interfaces, from rapid prototyping through controlled empirical evaluation. Published at CHI, UIST Adjunct, ETRA, MobileHCI, and GI.

RESEARCH EXPERIENCE

Research Associate

09 / 2025 – present

Ontario Tech University — Vialab

- Designed and deployed an interactive French-vocabulary learning system that uses real-world object identification to support contextual vocabulary learning.
- Contributing to a gaze-aware interaction study (ETRA 2026) that detects distraction and guides attention recovery during foreign-language video learning.

Undergraduate Student Research Fellow

09 / 2022 – 08 / 2023

Ontario Tech University — Vialab

- Designed and implemented GazeQ-GPT, a gaze-driven question-generation system for AI-augmented personalized learning from short educational videos.
- Ran the user study, analyzed quantitative and qualitative data, and first-authored the resulting GI 2025 paper.

Undergraduate Student Researcher

05 / 2022 – 08 / 2022

Ontario Tech University — Vialab

- Designed and implemented viaCross, a pen-based interaction technique for multi-object and multi-attribute selection using crossing gestures and in-place widgets.
- Designed and ran a controlled user study comparing viaCross with a traditional WIMP-style filtering interface; analyzed performance, accuracy, and user-experience trade-offs.

Research Assistant

07 / 2021 – 12 / 2021

Ontario Tech University

- Designed and deployed a web-based breadboard circuit simulator used to teach digital logic design.

SELECTED RESEARCH PROJECTS

AnnotateGPT — Pen & LLM Interaction

CHI 2026 · First author

- Turned handwritten pen annotations into an input modality for LLM agents, so reviewers can direct AI feedback with natural markup instead of switching to a chat interface.
- Designed a two-agent pipeline: one agent infers the purpose of each annotation, which the user confirms or corrects, and a second agent extends that feedback across the rest of the document.
- Ran a comparative study with 12 novice teachers against a non-AI pen baseline, showing how pen annotations let users regulate, refine, and adopt AI feedback.

NeuroSight — Eye-Tracking & BCI & Multimodal Interaction

UIST EA 2024 · First author

- Built a touch-free camera that combines eye-tracking glasses with a brain-computer interface, removing the need to hold or touch the device.
- Used gaze fixations to replace touchscreen selection and to infer context-aware camera modes, such as switching to translation when the user looks at foreign text.

SwipeSense — IMU Sensing & Novel Device Interaction

MobileHCI 2025 · Second author

- Enabled directional swipe input on the back of a smartphone using only its built-in IMU sensors and machine learning, with no extra hardware.
- Machine learning models distinguished back-of-device swipe directions with 72–95% accuracy, and the paper demonstrated example applications.

TECHNICAL SKILLS

Languages & Tools	Python, Java, C / C++, SQL, R, JavaScript / TypeScript, HTML / CSS, React, Git, PyTorch
HCI & UX Research	Controlled experiments, usability studies, mixed-methods, study design, statistical analysis, prototyping, A/B testing
Interaction & Sensing	Eye-tracking, brain-computer interfaces, IMU sensors, pen / stylus input, gaze-based interaction
AI & LLMs	LLM application development, prompt engineering, human-AI interaction design, AI-assisted software development
Prototyping	Rapid interactive prototyping, ideation, AI-assisted software development, sensor-integrated experimental systems, data-collection tooling

EDUCATION

Computer Science (MSc)

09 / 2023 – 08 / 2025

Ontario Tech University

- **Thesis:** Implicit Pen Annotation Assisted by Large Language Models — published at **CHI 2026** ([full thesis](#)).
- **Advisors:** Mariana Shimabukuro and Christopher Collins

Computer Science (BSc Hons), Minor in Mathematics

09 / 2017 – 05 / 2022

Ontario Tech University

- **Thesis:** Touch Free Camera: Mental Commands and Hand Gestures — published at **UIST EA 2024** ([full thesis](#)).
- **Advisor:** Christopher Collins
- President's List (Fall 2020 – Winter 2022); Dean's List (Fall 2017).

PUBLICATIONS

- [PP1] **Benedict Leung**, Mariana Shimabukuro, Christopher Collins. viaCross: A Pen-Based Technique for Selecting Objects and Attributes of Interest using Crossing-Based Gestures. **arXiv 2026**
- [C4] Mohammed Ahmed, **Benedict Leung**, Mariana Shimabukuro, Christopher Collins. Don't Wanna Miss a Thing: Gaze-Aware Implicit Interventions for Distraction Recovery in Foreign-Language Videos. **ETRA 2026**
- [C3] **Benedict Leung**, Mariana Shimabukuro, Christopher Collins. AnnotateGPT: Designing Human-AI Collaboration in Pen-Based Document Annotation. **CHI 2026**
- [C2] Neel Shah, **Benedict Leung**, Mariana Shimabukuro, Ali Neshati. SwipeSense: Exploring the Feasibility of Back-of-Device Swipe Interaction Using Built-In IMU Sensors. **MobileHCI 2025**
- [C1] **Benedict Leung**, Mariana Shimabukuro, Matthew Chan, Christopher Collins. GazeQ-GPT: Gaze-Driven Question Generation for Personalized Learning from Short Educational Videos. **Graphics Interface 2025**
- [EA1] **Benedict Leung**, Mariana Shimabukuro, Christopher Collins. NeuroSight: Combining Eye-Tracking and Brain-Computer Interfaces for Context-Aware Hand-Free Camera Interaction. **UIST EA 2024**

OTHER EMPLOYMENT

Teaching Assistant

09 / 2022 – 05 / 2025

Ontario Tech University

- Programming Workshop I & II · Software Systems Development & Integration · Human-Computer Interaction

HONOURS & AWARDS

NSERC USRA — \$8,400 · Competitive national award for 16 weeks of full-time research

05 / 2022

In-Course Scholarship · Awarded for a 4.0–4.3 GPA over the academic year

09 / 2020 – 09 / 2022